LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction using Kaggle Data
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1. Objective

To develop a machine learning classification model using Logistic Regression to predict diabetes outcomes.

- **Outcome = 0:** No diabetes
- **Outcome = 1:** Diabetes

2. Dataset

Dataset: Pima Indians Diabetes Database (Kaggle)

The dataset contains 768 records and 8 input features:

- Pregnancies
- Glucose
- BloodPressure
- SkinThickness
- Insulin
- BMI
- DiabetesPedigreeFunction
- Age

Target variable: Outcome

3. Model and Methodology

Algorithm: Logistic Regression with class balancing.

Preprocessing techniques:

- Replaced invalid zero measurements with missing values.
- Handled missing values using median imputation.
- Applied StandardScaler for feature scaling.
- Divided the dataset into training and testing sets (80:20).
- Trained the Logistic Regression model and generated predictions.

4. Evaluation Metrics

Metric	Result
Accuracy	0.734 (73.38%)
Precision	0.603 (60.32%)
Recall	0.704 (70.37%)
F1-score	0.650 (64.96%)
ROC-AUC	0.813

5. Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	TN = 75	FP = 25
Actual 1	FN = 16	TP = 38

6. Key Findings

The Logistic Regression model achieved a ROC-AUC score of approximately 0.813 and detected 70.37% of the positive cases in the test dataset.

The coefficient analysis showed that Glucose (+1.183) and BMI (+0.710) had the strongest positive coefficients among the input features.

Recall is an important metric in this classification task because false negatives represent actual positive cases incorrectly predicted as negative.

7. Limitations

The dataset is relatively small and represents a specific historical population. The model is developed for educational purposes and understanding machine learning classification. It is not intended for medical diagnosis.

8. Conclusion

Through this project, I learned how to build a Logistic Regression classification model using Python. I performed data cleaning, missing-value handling, feature scaling, model training and evaluation. The model achieved 73.38% accuracy and a ROC-AUC score of 0.813 on the test dataset. This project helped me understand how machine learning can be applied to classification problems and how different evaluation metrics measure model performance.